Mutual Fund Analytics Dashboard

BlueStocks Data Analytics Internship

Presented By:

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Introduction

- Mutual funds are one of the most popular investment options.
- Investors need data-driven insights to compare different schemes.
- This project analyzes mutual fund data and presents it through interactive dashboards.
- Technologies used include Python, SQLite, and Power BI.

Problem Statement

Large mutual fund datasets are difficult to analyze manually.
Investors require quick insights into NAV, AUM, SIP, and benchmark performance.
There is a need for an automated analytics dashboard.

project Objectives

- Build an ETL pipeline using Python.
- Clean and validate mutual fund datasets.
- Store data in SQLite.
- Perform exploratory data analysis.
- Create interactive Power BI dashboards.
- Generate meaningful financial insights.

Python Workflow

Modules developed:

- Data Ingestion
- Data Validation
- SQLite Loading
- Exploratory Data Analysis
- Risk Metrics Calculation

Libraries Used:

- Pandas
- NumPy
- SQLite3
- Matplotlib

Database Design

SQLite Database

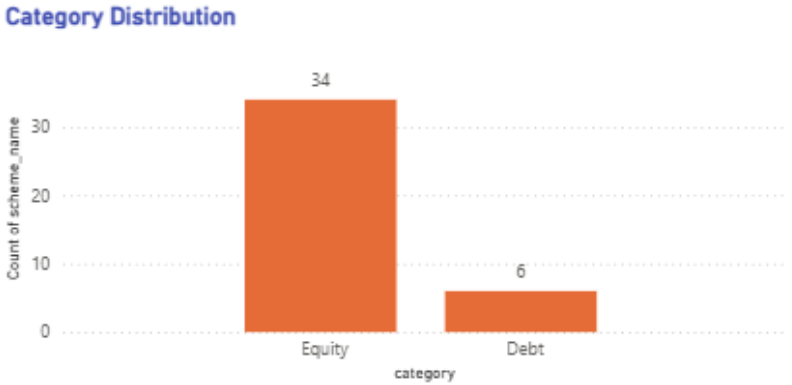
Tables Used:

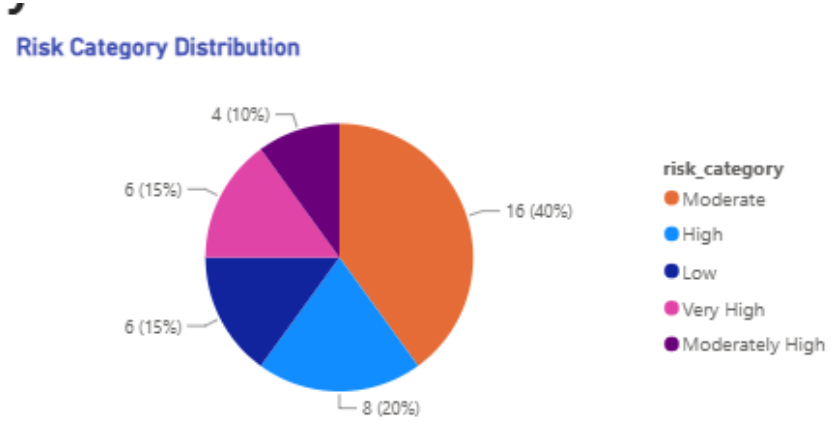
- Fund Master
- NAV History
- AUM by Fund House
- Monthly SIP
- Category Inflows
- Benchmark Indices

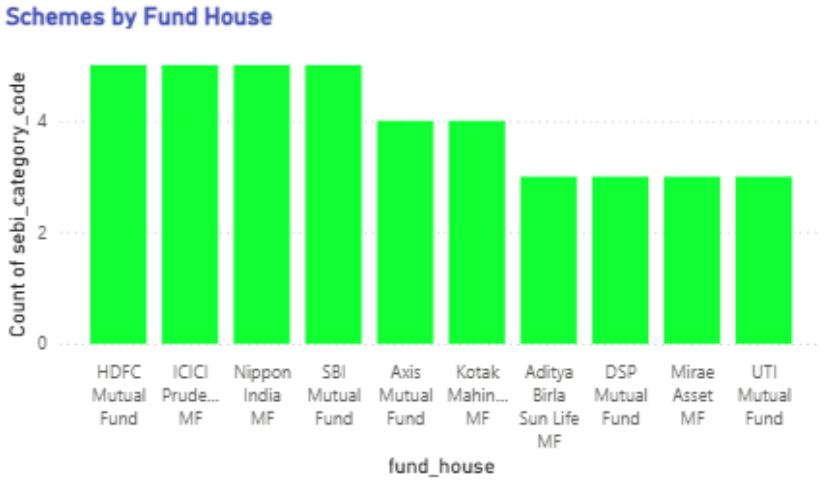
Purpose:

- Store cleaned data.
- Enable efficient querying.
- Support dashboard visualization.

Page 1 – Market Overview



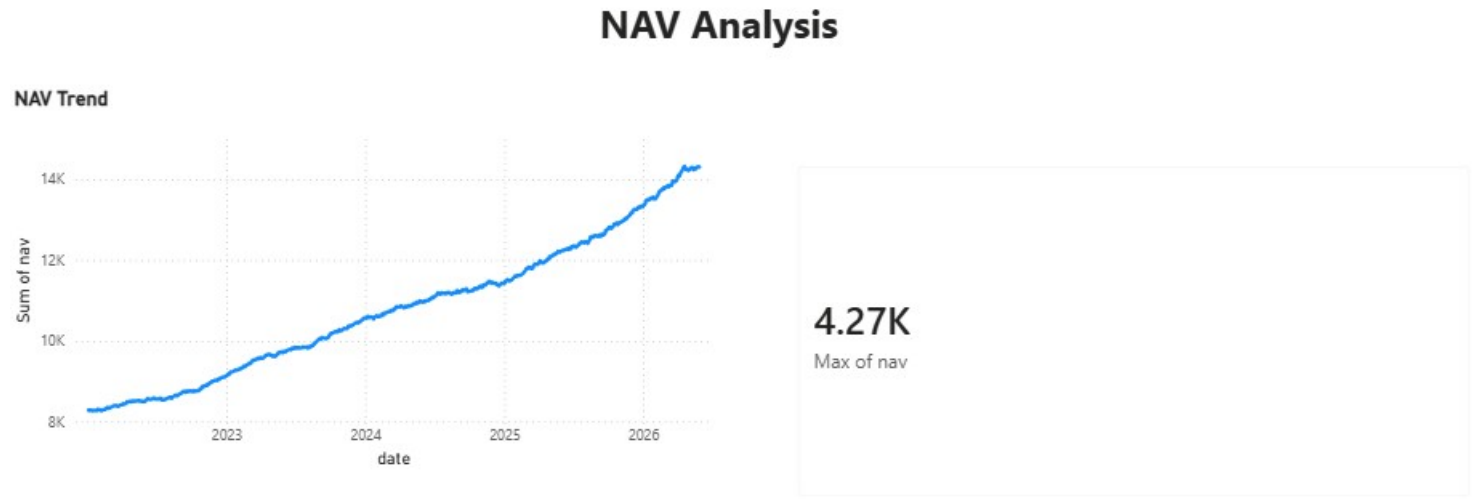




scheme_name	fund_house	category	risk_
UTI Nifty 50 Index Fund - Regular - Growth	UTI Mutual Fund	Equity	Mod
UTI Mid Cap Fund - Regular - Growth	UTI Mutual Fund	Equity	High
UTI Flexi Cap Fund - Regular - Growth	UTI Mutual Fund	Equity	Mod
SBI Small Cap Fund - Regular Plan - Growth	SBI Mutual Fund	Equity	Very
SBI Small Cap Fund - Direct Plan - Growth	SBI Mutual Fund	Equity	Very
SBI Magnum Gilt Fund - Regular Plan - Growth	SBI Mutual Fund	Debt	Low
SBI Bluechip Fund - Regular Plan - Growth	SBI Mutual Fund	Equity	Mod
SBI Bluechip Fund - Direct Plan - Growth	SBI Mutual Fund	Equity	Mod
Nippon India Small Cap Fund - Regular - Growth	Nippon India MF	Equity	Very
Nippon India Large Cap Fund - Regular - Growth	Nippon India MF	Equity	Mod
Nippon India Large Cap Fund - Direct - Growth	Nippon India MF	Equity	Mod
Nippon India Gilt Securities Fund - Regular - Growth	Nippon India MF	Debt	Low
Total			

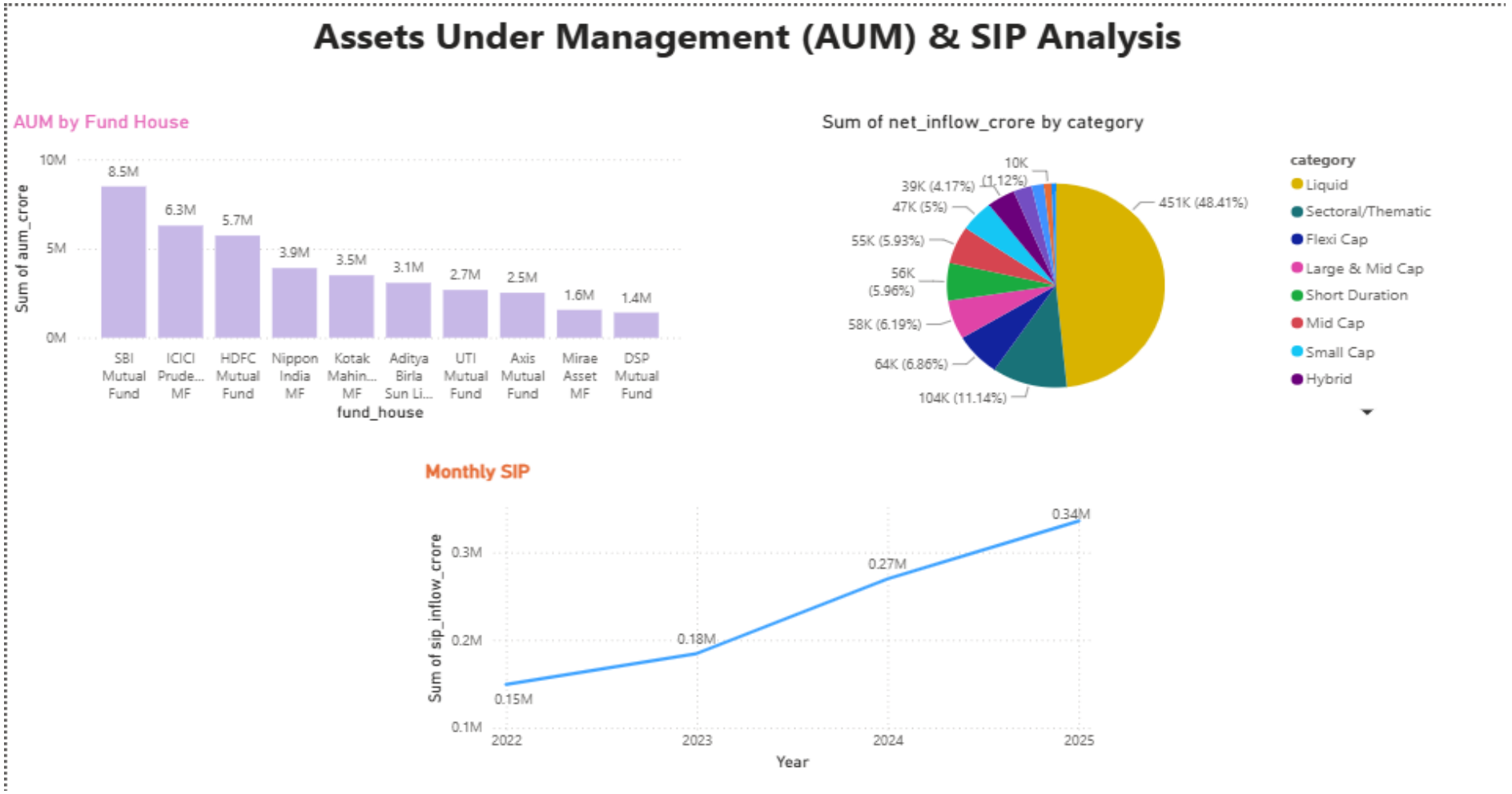
This page gives an overview of mutual funds. It shows fund categories, risk distribution, fund houses, and scheme details.

Page 2 – NAV Analysis



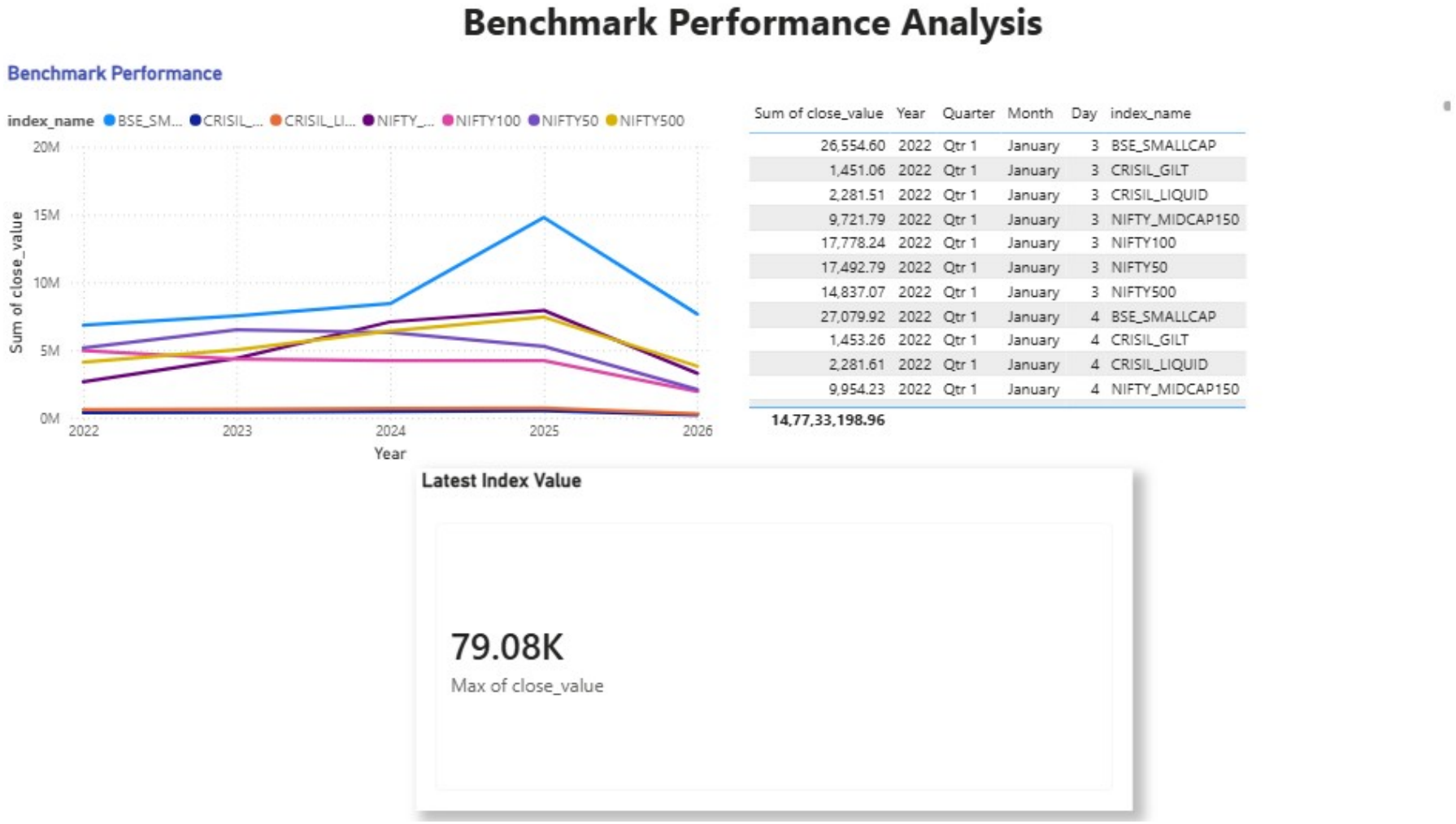
This page shows the NAV trend over time and the latest NAV value. It helps analyze mutual fund performance.

Page 3 – AUM & SIP Analysis



This page displays Assets Under Management, monthly SIP inflows, and category-wise inflows to understand investment trends.

Page 4 – Benchmark Analysis



This page compares benchmark performance over time and shows the latest benchmark value for market comparison.

Skills Acquired

Technical Skills:

- Python
- Pandas
- SQLite
- SQL
- Power BI

Professional Skills:

- Data Cleaning
- Data Visualization
- Problem Solving
- Dashboard Design
- Financial Analytics

Results

Key Outcomes

- Developed a complete ETL pipeline.
- Built an SQLite database.
- Created four interactive Power BI dashboards.
- Performed data analysis and risk metrics.
- Gained practical experience in financial data analytics.

Conclusion

- Successfully developed an end-to-end Mutual Fund Analytics Dashboard.
- Improved understanding of ETL, databases, and business intelligence.
- The project provides meaningful insights into mutual fund performance.

Thank you